University of Engineering and Technology Taxila

PROJECT REPORT

Anti-Sleep Glasses

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Abstract

The Arduino Anti-Sleep Glasses project is a smart wearable designed to detect drowsiness in drivers and students.

By using an eye-blink sensor, it provides timely alerts through an LED and a buzzer to prevent accidents and maintain focus.

This report details the development, implementation, and testing of the device.

Introduction

Drowsiness is a significant cause of road accidents and reduced productivity. This project aims to develop a wearable solution using Arduino technology to detect signs of fatigue and provide timely alerts.

It is designed to assist drivers and students in avoiding drowsiness-related issues.

Components

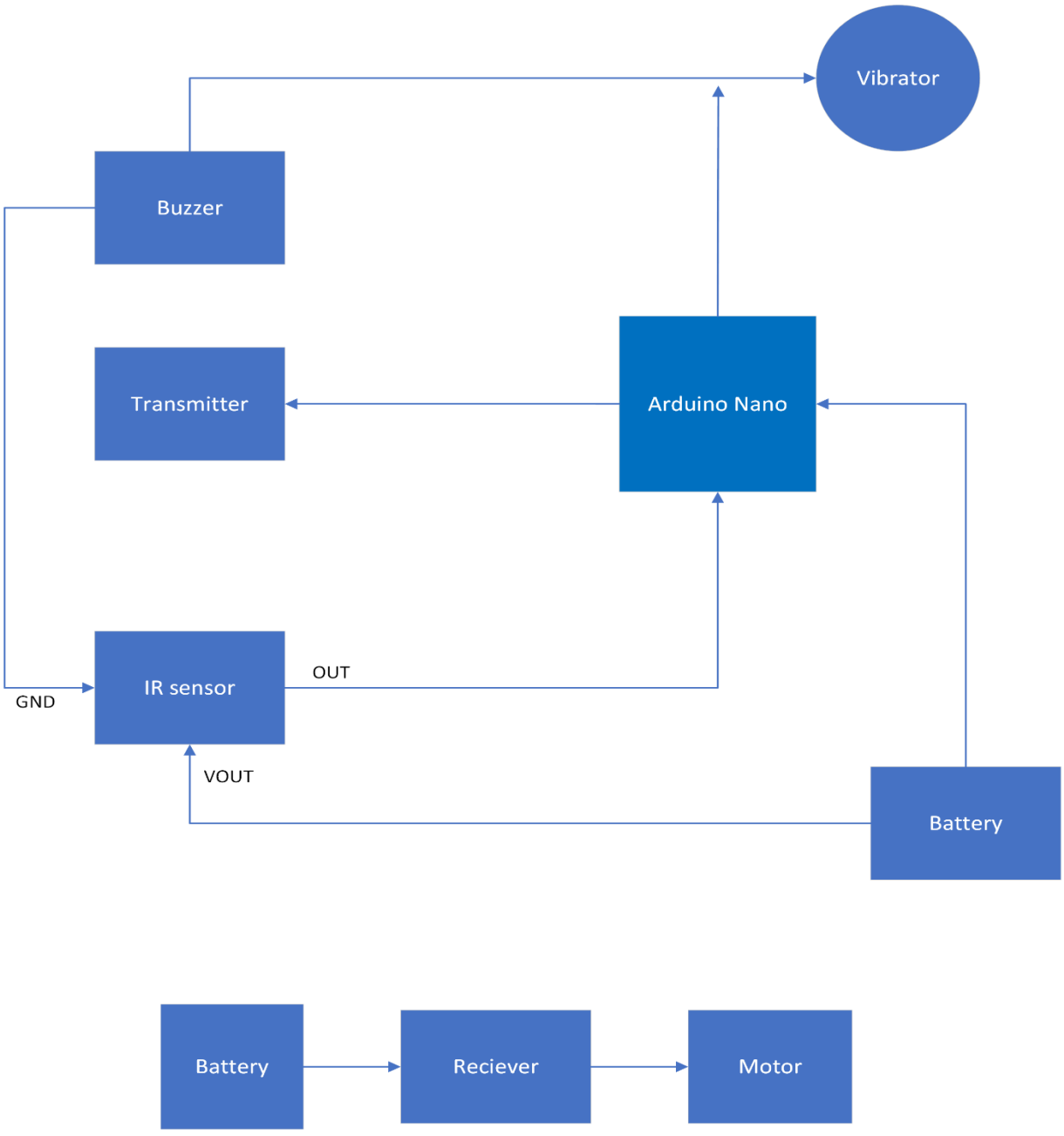
- Arduino Uno Board
- IR Sensor
- Relay
- Motor
- Buzzer
- LED Indicator
- Battery Pack
- Connecting Wires
- Breadboard

Methodology

The project involves connecting an eye-blink sensor to an Arduino board to monitor eyelid movement. Detected drowsiness triggers the Arduino to activate a buzzer and LED, waking the user.

The system was prototyped and tested on a breadboard before finalizing the design on a wearable frame.

Block Diagram

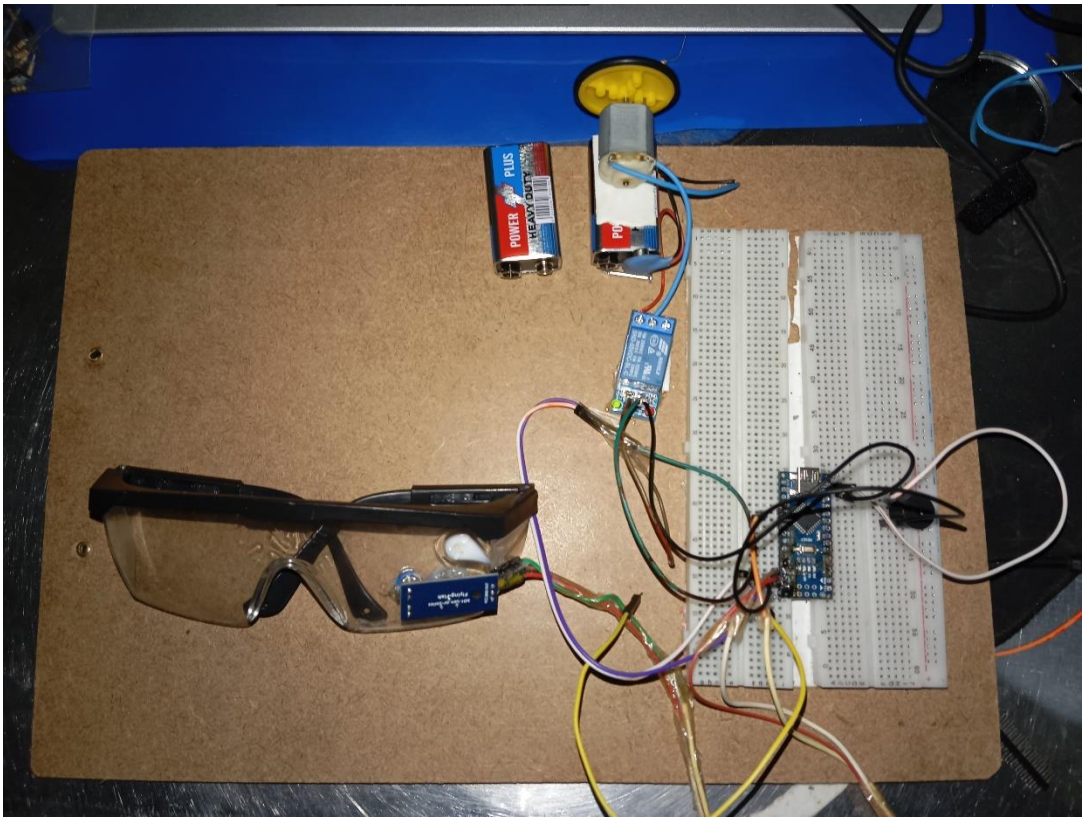


Implementation

The circuit was assembled on a breadboard using the listed components. The Arduino was programmed to process sensor input and activate outputs accordingly. The final design was mounted onto glasses for user testing.

Testing and Results

- The glasses successfully detected reduced blinking patterns simulating drowsiness.
- The buzzer and LED provided effective alerts in real-time.
- Outcome Picture:



Conclusion

The Arduino Anti-Sleep Glasses effectively address drowsiness detection in drivers and students. The project demonstrates the potential of simple, low-cost electronics in solving critical safety challenges. Future iterations could focus on improving the design's comfort and accuracy.